

Yen Lin (Darren) Chao

Contact



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Education

Master of Arts in Design
Engineering

**Brown University
and**

**Rhode Island School
of Design (RISD)** 2025-Present

Bachelor of Science in
Electrical Engineering
and Computer Science
(EECS) 2021-2024

**University of
California, Berkeley**

Skills

Design & Fabrication

Autodesk Fusion 360, Adobe
Illustrator, Laser Cutting, 3D
Printing/Extrusion, Rapid
Prototyping, Rhino, Sheet Metal
Working, Ceramic Firing,
Embroidery

Languages

Python, Java, C, C++, Go, SQL,
RISC-V, HTML, CSS, Javascript

Cloud & DevOps

Azure IoT Hub, Jenkins, Maven, Git,
Jira, Particle Cloud



Work Experience

Recycle PLA Researcher

Brown Design Workshop

August 2025 - Present

- Architected an end-to-end IoT monitoring system using Particle microcontrollers and sensors to track critical environmental variables (humidity/temperature) during PLA processing.
- Engineered data pipelines by routing real-time sensor data to Azure IoT Hub, automating the flow into a custom SQL database for centralized storage.
- Optimized a closed-loop recycling workflow, managing the transition of PLA waste from pelletizing and grinding to industrial extrusion and heat-pressing.
- Conducted rigorous material testing, including the calibration of laser cutter settings and heat-press parameters (oven/T-shirt press) to ensure structural integrity of upcycled products.
- Led physical product design initiatives, transforming recycled PLA waste into retail-ready consumer goods sold at the Brown University Bookstore.
- Facilitated stakeholder meetings with retail partners and logistics teams to align fabrication capabilities with market demand and pricing strategies.

Pottery Studio Instructor

Brown Design Workshop

January 2026 - Present

- Led weekly small group ceramics classes with hands on instruction and individualized guidance, fostering a supportive and collaborative learning space.
- Managed studio operations including organizing kiln firings, transporting work to off site facilities, maintaining equipment and materials, and coordinating with instructors to ensure smooth studio workflow

Software Engineer Intern

Development Bank of Singapore - Taipei

Summer 2024

- Collaborated within an Agile team, following Scrum methodologies with weekly stand-up meetings and sprints.
- Designed and implemented RESTful APIs for the "GiftBox" campaign on the CARD+ app and manipulated XML for form structures and data, directly interacting with the database and processing form inputs.
- Enhanced unit test coverage for legacy APIs to 100%, utilizing mocking and spy frameworks, and Cucumber.
- Deployed and tested APIs using Jenkins, monitored logs for debugging and validation, and managed code using Maven, Git, and Jira.



Coding Projects (see darrenchao.com for design projects)

PACMAN

Modeling Multi-agent Game with ML

February 2024

- Utilized Value Function, Q learning, and Deep Q Network algorithms to enhance PACMAN's gameplay.
- Implemented alpha-beta tree pruning algorithm to efficiently search for and select the optimal path for PACMAN.

REGULUS

Secure File Sharing System

November 2023

- Crafted a multifaceted security infrastructure for the file sharing system, integrating robust user verification mechanisms to authenticate users and stringent file modification protocols to ensure data integrity.
- Employed cryptographic tools including RSA for public-key encryption, HMACs for message authentication, and symmetric key encryption for secure data transmission.
- Established secure communication channels utilizing protocols like TLS/SSL to protect data in transit and mitigate risks associated with interception or eavesdropping.

Build Your Own World

Generating Random Explorable and Interactive Worlds March 2023

- Architected an algorithm for generating pseudo-random worlds with customizable seeds in Java.
- Implemented a modified Kruskal's algorithm to connect randomly generated rooms with hallways efficiently.
- Currently implemented features include: pseudo-random world generation with connected rooms, the ability to start a new game, load game, or replay last save, light sources, different levels of the game, and locked rooms.